

RAHULSAI SINGIREDDY

DATA ENGINEER

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SUMMARY

Data Engineer with **4+ years** of experience designing and optimizing **data pipelines, cloud-native architectures**, and **analytics ecosystems** in **Healthcare** and **Finance** domains. Skilled in **Python, SQL, Apache Spark**, and **Azure Data Factory** to modernize **ETL workflows**, enhance dataset reliability, and enable scalable analytics. Experienced in building high-performance data models using **Snowflake, MongoDB, and AWS Redshift**, improving query efficiency and reporting consistency. Delivered actionable insights through **Power BI** and **Tableau** dashboards, integrating ML outputs, advanced Excel, and **A/B testing** to support data-driven decision-making across enterprise environments.

TECHNICAL SKILLS

Programming Language:	Python, SQL, Java, R
Data Processing & Analytics:	Pandas, NumPy, PySpark, Dask, Apache Spark
Machine Learning & AI:	Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, LightGBM
Data Engineering & ETL:	Apache Airflow, Azure Data Factory, Talend, Informatica
Databases & Data	PostgreSQL, MySQL, Snowflake, Redshift, BigQuery, MongoDB
Data Engineering Concept:	Apache Spark, Apache Hadoop, Apache Kafka, Apache Beam, ETL/ELT, MySQL, PySpark
Visualization & BI Tools:	Tableau, Power BI, Looker, Matplotlib, Seaborn
Cloud Platforms:	AWS (S3, EC2, SageMaker), Azure (Blob Storage, Synapse), GCP (BigQuery, AI Platform)]
Containerization & Deployment:	Docker, Kubernetes (basic), MLflow
Version Control Tools:	Git, GitHub, GitLab, Jira

EXPERIENCE

UnitedHealth Group | Data Engineer | IN, USA

May 2023– Present

- Built and maintained HIPAA-compliant data ingestion pipelines using **Apache NiFi and Python** to process high-volume **EHR/EMR, claims** and **provider data**, improving downstream data reliability and reporting accuracy.
- Designed and optimized **MongoDB** collections with schema validation and indexing for clinical and operational datasets, reducing query latency by **25%** and enabling faster analytics for care-management teams.
- Developed and monitored **Apache Airflow DAGs** to orchestrate multi-source healthcare **ETL workflows**, automating daily refreshes across claims and encounter data and achieving **99% pipeline** success rates.
- Implemented scalable **Apache Spark (PySpark)** transformations integrated with **AWS Redshift**, processing millions of records per run and reducing data availability latency by 22% for analytics consumers.
- Collaborated with analytics and operations teams to deliver curated datasets and Tableau-ready data models, improving KPI visibility for patient throughput and claims operations and shortening decision cycles by 20%.
- Applied lightweight forecasting and validation logic using **Python** and **SQL** to support utilization trend analysis, contributing to more accurate capacity planning and operational forecasting.

Goldman Sachs | Data Engineer | India

Mar 2020 – Dec 2021

- Engineered end-to-end **Azure Data Factory (ADF) pipelines** to ingest and transform large-scale **financial transactions, credit, and loan risk data** from SQL Server sources, improving pipeline stability and reducing recurring failures.
- Created and enhanced **Snowflake data models (star schemas)** for **customer accounts, credit exposure, delinquency, and repayment datasets** using advanced SQL, enhancing query performance by 26% and supporting **audit-ready regulatory reporting**.
- Developed scalable **PySpark transformations on Databricks** to process **high-volume transactional and credit datasets**, ensuring consistent delivery of **model-ready data** for **risk, compliance, and analytics teams**.
- Applied **CI/CD pipelines and automated deployment** on AKS, reducing manual release effort and improving ETL reliability for finance-specific environments.
- Integrated **external regulatory, credit bureau, and market data APIs** into enterprise pipelines, improving data reconciliation and completeness of **credit-risk and regulatory reporting datasets**.
- Delivered optimized **Power BI semantic models and dashboards** for credit-risk, loan performance, and portfolio analysis, reducing analyst data-prep time and accelerating business review cycles.

PROJECTS

Healthcare Claims Anomaly Detection & Cost Leakage Analysis

- Built scalable **PySpark pipelines** to clean, normalize, and aggregate multi-year claims data (8M records), reducing data processing time by **35%**.
- Engineered cost and utilization features using **SQL and Python**, enabling detection of abnormal billing, duplicate claims, and provider-level outliers.
- Trained and evaluated **Isolation Forest and XGBoost models** to flag anomalous claim patterns, achieving **82% precision** on validation datasets.
- Presented curated datasets and **Tableau dashboards** to visualize cost leakage trends, supporting faster identification of high-risk providers and services.

Credit Risk Probability & Customer Default Prediction Platform

- Developed a modular **data preprocessing framework** using **Python and SQL**, handling missing values, imbalance, and categorical encoding at scale.
- Built and improved **LightGBM and Logistic Regression models** to predict probability of default, improving AUC from **0.71 to 0.84** through feature engineering.
- Implemented batch scoring pipelines on **Databricks**, generating customer-level risk scores and segmenting portfolios by exposure and delinquency risk.
- Developed **Power BI dashboards** to track portfolio risk distribution and scenario outcomes, reducing manual analyst effort

CERTIFICATIONS

- Snowpro Advanced Data Engineer
- Databricks Certified Data Engineer Professional

EDUCATION

Master of Science in Engineering Management and Information Systems - Indiana Institute of Technology, Indiana, USA

Bachelor of Technology - Anurag University, Hyderabad, Telangana, India